

FUJITSU MEGA HM-400 Brush Cutter

Test Report Number	:	2013 - 118
Date of Test	:	September 19, 2013
Test Valid Until	:	September 19, 2018
Brand	:	FUJITSU MEGA
Model	:	HM-400
Serial Number	:	No Data
Type	:	Portable, sling, straight shaft
Manufacturer	:	Made in China
Test Requested by	:	M & W DISTRIBUTING CO., INC. 14 Panay Avenue, Quezon City
Machine Classification	:	Commercial

SUMMARY

The FUJITSU MEGA HM-400 Brush Cutter was tested for field performance using a three-bladed steel cutting attachment. It had an Actual Field Capacity of 0.095 ha/h at an Engine Speed of 3943 rpm. It had a Swath of 2.45 m and Noise Level of 88.9 db (A). The 1.49 kW FUJITSU gasoline engine had a Fuel Consumption Rate of 1.02 L/h.

The Brush Cutter operates smoothly and no breakdown was observed.

SCOPE OF TEST

The test was conducted on September 19, 2013 at the Agricultural Machinery Testing and Evaluation Center (AMTEC), College of Engineering and Agro-industrial Technology, University of the Philippines Los Baños, College, Laguna. It was tested to determine the following:

1. Verification of manufacturer's specifications
2. Actual field performance

METHOD OF TEST

The FUJITSU MEGA HM-400 Brush Cutter was evaluated based on AMTEC Test Standard for Brush Cutters.

OPERATOR'S MANUAL

No operator's manual was submitted with the brush cutter.

DESCRIPTION OF THE BRUSH CUTTER

The FUJITSU MEGA HM-400 Brush Cutter was a portable, sling type, straight shaft (**Figure 1**) consisting of the following main components: cutting assembly, handlebars and controls, harness with side guard, and the prime mover.

The cutting assembly consisted of the drive shaft, deflector, and the cutting attachment. The engine powered the cutting attachment through the steel drive shaft (**Figure 2a**). The deflector (**Figure 2b**), which acted as the protective cover of the cutting attachment, was made of plastic and could accommodate cutting attachments with a maximum diameter of 430 mm. The two-bladed steel cutting attachment used during the actual field testing had a cutting length of 310 mm (**Figure 3**).

The handles (**Figure 4a**) of the brush cutter, where the throttle lever was also located, were covered with protective rubber material. The throttle lever (**Figure 4b**) was located at the right handlebar.

The brush cutter was used with a harness (**Figure 4c**) for ease and safety of operation. Some parts of the harness were padded for the comfort of the operator.

The brush cutter primemover (**Figure 5**) was a FUJITSU two-stroke cycle, air-cooled, single-cylinder gasoline engine. The engine was started by a rope recoil system. The plastic fuel tank was located at the bottom of the engine.

The detailed specifications of the brush cutter are listed in **Table 1**.

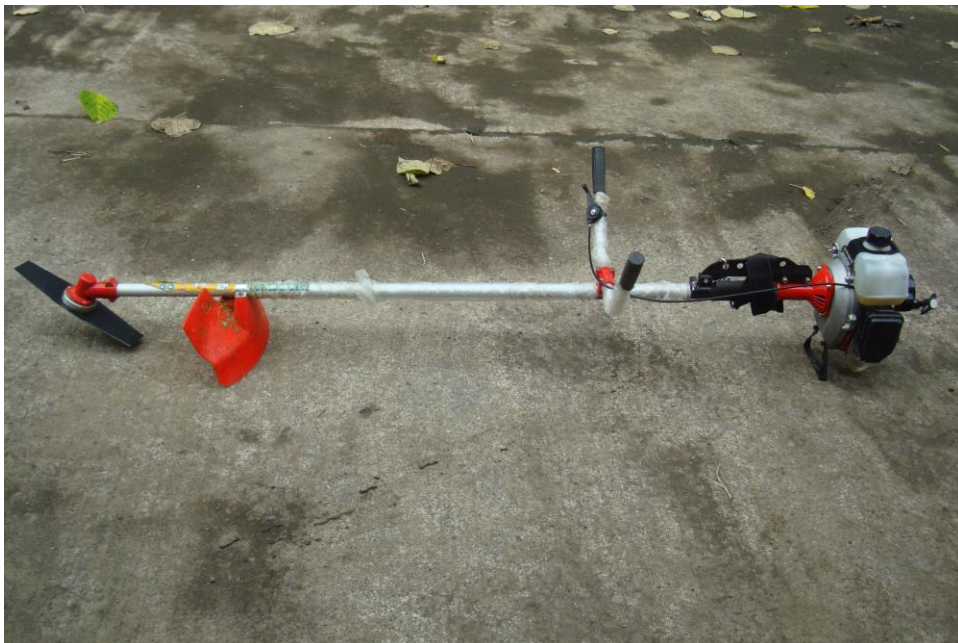


Figure 1. FUJITSU MEGA HM-400 Brush Cutter.

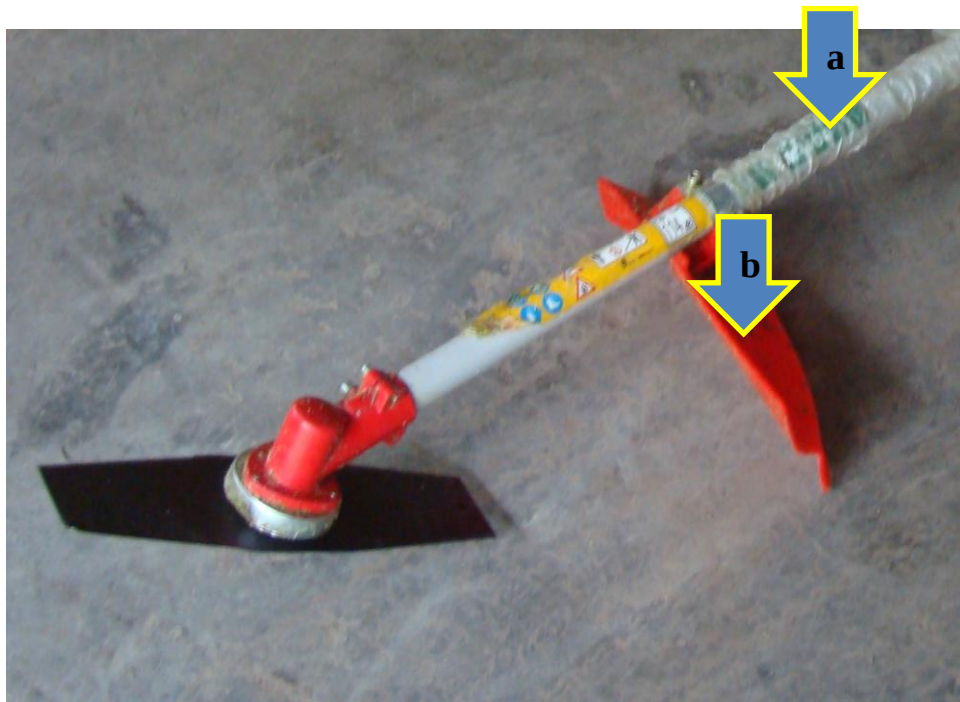


Figure 2. Drive shaft (a) and deflector (b)



Figure 3. Cutting blade used during the actual field testing

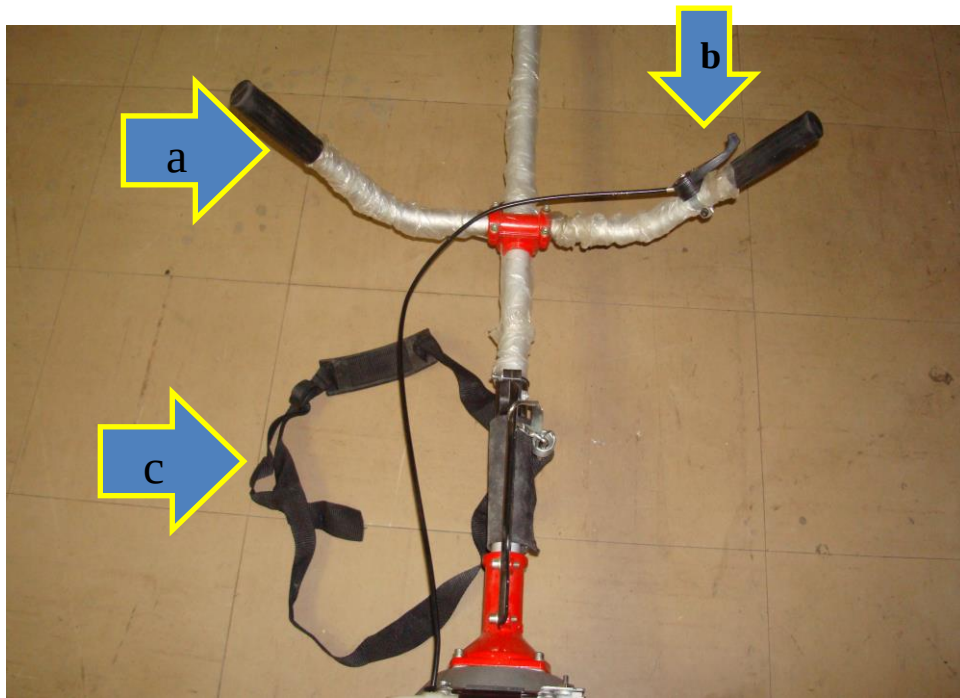


Figure 4. Handlebar (a), Throttle Lever (b) and Harness (c)



Figure 5. Engine

SPECIFICATIONS

Table 1. Detailed specifications of the FUJITSU MEGA HM-400 Brush Cutter.

1. Brush Cutter			
1.1	Brand	:	FUJITSU MEGA
1.2	Model	:	HM-400
1.3	Serial Number	:	No Data
1.4	Make	:	China
1.5	Type	:	Portable, sling, straight shaft
1.6	Overall Specifications		
1.6.1	Length, mm	:	1930
1.6.2	Width, mm	:	580
1.6.3	Height, mm	:	410
1.6.4	Weight, kg	:	8.50
1.7	Cutting blade		
1.7.1	Cutting length, mm	:	310
1.7.2	Material	:	Steel
1.7.3	Configuration	:	Two-bladed
1.8	Deflector		
1.8.1	Length, mm	:	219
1.8.2	Height, mm	:	200

Table 1. Detailed specifications of the FUJITSU MEGA HM-400 Brush Cutter (cont.).

2. Engine			
2.1	Brand	:	FUJITSU
2.2	Type	:	Two-stroke, air -cooled, single cylinder gasoline engine
2.3	Model	:	HM-400
2.4	Make	:	China
2.5	Serial Number	:	No data available
2.6	Rated Output Power (kW)	:	1.49 kW
2.7	Number of Cylinder	:	1
2.8	Total displacement volume (cc)	:	No data available
2.9	Bore, mm	:	No data available
2.10	Stroke, mm	:	No data available
2.9	Fuel system		
	2.9.1 Type of fuel feed	:	By gravity
	2.9.2 Fuel used	:	Gasoline
2.10	Cooling system	:	Air-cooled
2.11	Starting system	:	Rope recoil
2.12	Exhaust system	:	Muffler

RESULTS AND DISCUSSION

Test Site

The FUJITSU MEGA HM-400 Brush Cutter was tested at the AMTEC Test Field. The density of the grass was high as shown in **Figure 6**. The Average height of grasses was 1729 mm.

Field performance test data are shown in **Table 2**.

Performance

The Performance of the brush cutter was evaluated in terms of its Field Efficiency and Quality of work.

Actual Field Capacity. It is the ratio of the area where the grasses are cut and the total time of operation. The Field Capacity of the machine was 0.095 ha/h or 10.5 h/ha.

Field Efficiency. It is the ratio of the Actual Field Capacity and the Theoretical Field Capacity, expressed in percent. The Field Efficiency of the machine was 86.8%.

Quality of work. The quality of work of a brush cutter was determined in terms of the percentage of grass cut. Based on ocular inspection the brush cutter had totally cut all the grasses on the test lot (**Figure 7**).

Noise Level. The Noise Level was obtained at 50 mm away from the ear of the operator. The Average Noise Level measured was 88.9 db(A).

Fuel Consumption. The Fuel Consumption of the 1.49 kW two-stroke gasoline engine was 1.02 L/h.

Operational Pattern. The Brush cutter was tested using continuous pattern turn strips each end.

PERFORMANCE TEST

Brush cutter on test : FUJITSU MEGA
 Model : HM-400
 Date of test : September 19, 2013
 Location of test : AMTEC, CEAT, UPLB, College, Laguna

Table 2. Field Performance test of the FUJITSU MEGA HM-400 Brush Cutter

<u>Parameters</u>	<u>Data</u>
1. Operation	Grass cutting
2. Grass density	High
3. Average grass height (mm)	1729
4. Field test area (m ²)	165.5
5. Total time of operation (h)	0.17
6. Speed of Operation (kph)	0.446
7. Average width of cut (mm)	245
8. Noise Level [db(A)]	88.9
9. Fuel Consumption (1.49 kW gasoline engine) (L/h)	1.02
10. Blade rotation during operation (rpm)	3943
11. Actual Field Capacity ha/h h/ha	0.095 10.5
12. Theoretical Field Capacity ha/h h/ha	0.109 9.15
13. Field Efficiency (%)	86.8



Figure 6. The grass on the test site during testing



Figure 7. Quality of work

Test Engineers:

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N.B.

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